

Program -3:

Code a dockerized python flask or [node.js](#) application

Program structure:

Dockerfile

[app.py](#)

Requirements.txt

Step 1: create a Dockerfile

FROM python:latest

WORKDIR /app

COPY requirement.txt .

RUN pip install --no-cache-dir -r requirement.txt

COPY . .

EXPOSE 5000

CMD ["flask", "run"]

Step 2: create a [app.py](#)

```
from flask import Flask
```

```
app=Flask(__name__)
```

```
@app.route("/")
```

```
def hello():
```

```
    return "hello from the simple flask application"
```

```
if __name__=="__main__":
```

```
    app.run(host="0.0.0.0",port=5000)
```

Step 3: requirements.txt

```
Flask==2.3.3
```

Step 4:build the image

```
docker build -t program-3 .
```

Step 5: run the container

```
docker run -p 5000:5000 program-3
```

Step 6:verify

<http://localhost:5000>

```

42aeba89be41    secondfile    "redis-server"    9 days ago    Exited (0) 14 hours ago
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg3$ ls
app.py Dockerfile requirements.txt
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg3$ docker build -t flask .
[+] Building 2.1s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 200B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b680f
=> [internal] load build context
=> => transferring context: 93B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:69ae3ee2b53e6b4d3ec269662b21dd9d84b812b965cbb47df78f8613556efbe3
=> => naming to docker.io/library/flask
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg3$ docker run -p 5000:5000 flask
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
^C1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/prg3$ cd ..
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles$ ls

```